Suggested problems: 2-4, 6-10

Problem 1: Insurance companies keep track of how likely various cars are to be stolen. Suppose that a company in a particular area computes the joint distribution of car brands and the indicator of whether the car will be stolen during a particular year that appears in the table below.

	Brand Y					
Stolen X	1	2	3	4	5	Total
0	0.129	0.298	0.161	0.280	0.108	0.976
1	0.010	0.010	0.001	0.002	0.001	0.024
Total	0.139	0.308	0.162	0.282	0.109	1.000

Compute the conditional probability function $g_{X|Y}(x \mid y)$. Which brand of car is most likely to be stolen?

Problem 2: A manufacturing process consists of two stages. The first stage takes Y minutes, and the whole process takes X minutes (which includes the first Y minutes). Suppose that X and Y have a joint continuous distribution with probability density function

$$f(x,y) = \begin{cases} e^{-x} & : 0 \leq y \leq x, \\ 0 & : \text{else.} \end{cases}$$

Compute the conditional probability density function $g_{X|Y}(x \mid y)$.